a superbomb effort. The slave labor of zeks, the theoretical research facilities, the manufacturing facilities, the test facilities, the multiple teams of physicists on each aspect of the design and construction, all must be focused on trying to beat the Americans to the hydrogen bomb. Of this the Americans, in the midst of their debate over whether to mount a crash effort on the super, knew nothing. However, the Americans had superior technology and a large head start.

On 1 November 1952, the Americans exploded a hydrogen bomb–type device code-named *Mike.* Mike was designed to test the 1951 Teller–Ulam invention and was based on the design computations of Wheeler's team and tile cooperating team at Los Alamos. It used liquid deuterium as its principal fuel. To liquify the deuterium and pipe it into the explosion region required an enormous, factory-like apparatus. Thus, this was not the kind of bomb that one could deliver on any airplane or rocket. Nevertheless, it totally destroyed the island of Elugelab in the Eniwetok Atoll in the Pacific Ocean; it was 800 times more powerful than the bomb that killed over 100,000 people in Hiroshima.

On 5 March 1953, amidst somber music, Radio Moscow announced that Joseph Stalin had died. There was rejoicing in America, and grief in the U.S.S.R. Andrei Sakharov wrote to his wife, Klava, "I am under the influence of a great man's death. I am thinking of his humanity."

On 12 August 1953, at Semipalatinsk, the Soviets exploded their first hydrogen bomb. Dubbed *Joe-4* by the Americans, it used Sakharov's layered-cake design and Ginzburg's LiD fusion fuel, and it was small enough to deliver in an airplane. However, the fuel in Joe-4 was *not* ignited by the Teller–Ulam method, and as a result Joe-4 was rather less powerful than the Americans' Mike: "only" about 30 Hiroshimas, compared to Mike's 800.

In fact, in the language of the American bomb design physicists, Joe-4 was not a hydrogen bomb at all; it was a *boosted atomic bomb,* that is, an atomic bomb whose power is boosted by the inclusion of some fusion fuel. Such boosted atomic bombs were already part of the American arsenal, and the Americans refused to regard them as hydrogen bombs because their layered-cake design did not enable them to ignite an *arbitrarily large* amount of fusion fuel. There was no way by this design to make, for example, a "doomsday weapon" thousands of times more powerful than Hiroshima.